

**Home Conceptual Laboratory Activity 6****Procedure:**

Explore the simulation. Use the given link. Increase and decrease the length, area (cross-sectional area), and resistivity. Make observations

**Table:**

| Resistivity                                         | Resistance    | Length | Resistance   | Area                | Resistance   |
|-----------------------------------------------------|---------------|--------|--------------|---------------------|--------------|
| Copper $1.7 \times 10^{-8} \Omega \cdot \text{m}$   | 2.0 $\Omega$  | 0.5 m  | 1.0 $\Omega$ | 5 mm <sup>2</sup>   | 0.5 $\Omega$ |
| Nichrome $1.1 \times 10^{-6} \Omega \cdot \text{m}$ | 120 $\Omega$  | 1.0 m  | 2.0 $\Omega$ | 2.5 mm <sup>2</sup> | 1.0 $\Omega$ |
| Carbon $3.5 \times 10^{-5} \Omega \cdot \text{m}$   | 3800 $\Omega$ | 2.0 m  | 4.0 $\Omega$ | 1 mm <sup>2</sup>   | 2.0 $\Omega$ |
| Silver $1.6 \times 10^{-8} \Omega \cdot \text{m}$   | 1.8 $\Omega$  | 0.25 m | 0.5 $\Omega$ | 10 mm <sup>2</sup>  | 0.2 $\Omega$ |
| Iron $1.0 \times 10^{-7} \Omega \cdot \text{m}$     | 15 $\Omega$   | 1.5 m  | 3.0 $\Omega$ | 1.5 mm <sup>2</sup> | 1.5 $\Omega$ |

**Observations:**

- Longer wire = higher resistance.
- Thicker wire = lower resistance.
- Higher resistivity = higher resistance.
- Good conductors (copper, silver) have low resistance.
- Poor conductors (carbon) have very high resistance.

**Question.**

How will you minimize the resistance in a conducting wire?

Use a short, thick wire made of a low-resistivity material like copper or silver.

**Documentation:**